Autodesk®
Scaleform®

Scaleform

DirectX 9 Scaleform 3D

Ben Mowery

3.04

2011 8 4

Autodesk®
GAMEWARE

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Autodesk® Scaleform® 4.2

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1 ..... **1**
2 ..... **2**
3 组建安装和关联编译 **3**
3.1 3
3.2 4
3.3 Scaleform Player SWF 4
3.4 6
3.5 Scaleform 7
3.5.1 AS3_Global.h and Obj\AS3_Obj_Global.xxx **8**
4 ..... **10**
4.1 Flash 10
4.2 18
4.3 19
4.3.1 ..... **19**
4.3.2 ..... **20**
4.3.3 点击测试 **22**
4.3.4 ..... **23**
5 ..... **24**
5.1 24
6 C++、Flash **27**
6.1 C++ 27
6.1.1 FSCommand **27**
6.1.2 ExternalInterface **30**
6.2 C++ 32
6.2.1 ..... **32**
6.2.2 ..... **35**
6.3 多 Flash 文件之间通信 36
7 渲染到纹理 **42**

8 GFxExport 44
9 45

1

Scaleform UI
Adobe Flash Studio GPU
Scaleform Flash Studio

UI Scaleform 3D Flash Flash
3D 3D Flash UI Scaleform
UI

Scaleform Flash
DirectX ShadowVolume SDK

Scaleform Scaleform
Scaleform
Scaleform

Scaleform Scaleform 4.0

DirectX
SDK ShadowVolume Scaleform “cannot create
renderer” Scaleform Player

2

Scaleform Scaleform
https://gameware.autodesk.com/scaleform/developer/

- Web Scaleform SDK
- PDF
- Scaleform C++ API

- XML Scaleform XML
- Scale9 Grid Scale9Grid
- IME Scaleform IME Flash
  IME
- Scaleform

3

3.1

Windows Scaleform

Scaleform installer DirectX Scaleform C:\Program Files\Scaleform\GFx SDK 4.1

3rdParty

Scaleform libjpeg zlib

Projects

Visual Studio

App\Samples

Apps\Tutorial

Bin

Flash

FxPlayer: Flash HUD

IME: IME Flash

Samples: Flash FLA

Video Demo Scaleform

Win32: Scaleform Player

AMP: AMP Flash

GFxExport: GFxExport Flash

7

Include

Scaleform

Lib

Scaleform

Resources

CLIK

Doc

2 PDF

3.2

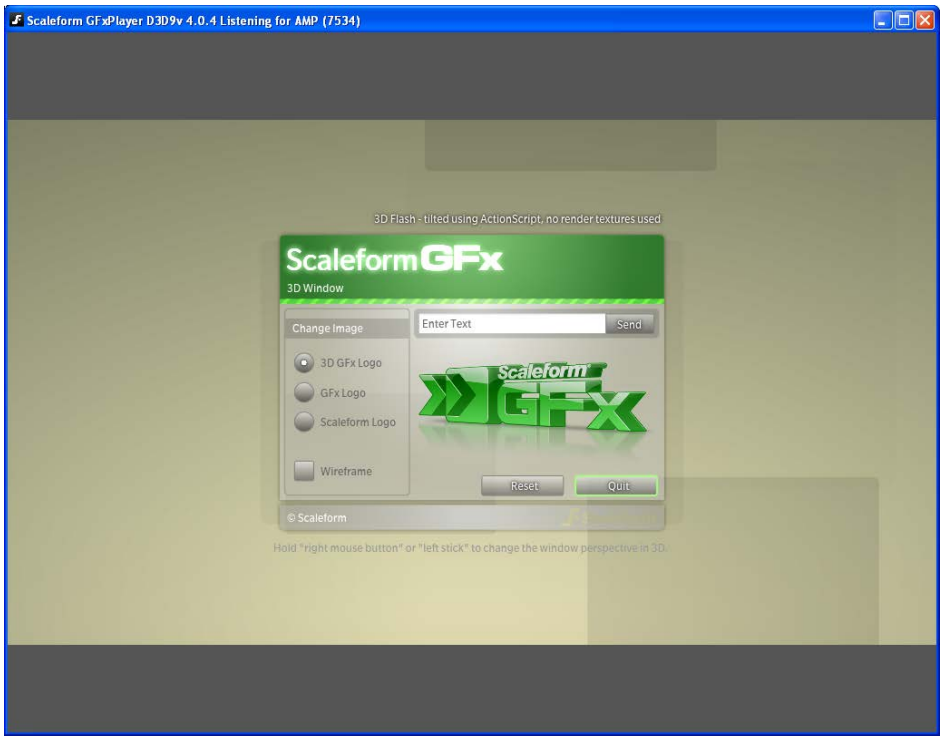
Scaleform C:\Program Files (x86)\Scaleform\GFx SDK 4.1\Projects\Win32\Msvc90\Demos\GFx 4.1 Demos.sln Demos.sln Visual Studio .sln D3D9_Debug_Static Scaleform Player

GFx SDK 4.1\Bin\Win32\Msvc90\GFxPlayer\GFxPlayer_D3D9_Debug_Static.exe

Start → All Programs → Scaleform → GFx SDK 4.1 → Demo GFx Player D3Dx SWF Scaleform Flash

3.3 Scaleform Player SWF

Start → All Programs → Scaleform → GFx SDK 4.1 → GFx Players→ GFx Player D3D9 C:\Program Files\Scaleform\GFx SDK 4.1\Bin\Data\AS2\Samples\SWFToTexture 3DWindow.swf



1 Flash

F1

1. CTRL+F FPS
2. CTRL+W Scaleform Flash
   CTRL+W
3. CTRL
4. CTRL
5. CTRL+Z
6. CTRL+U
7. F2

Flash

800x600 1600x1200

Scaleform “Scaleform”
Flash Scaleform

“3D Scaleform Logo” CTRL+W
CTRL+A EdgeAA

FSAA

AA Scaleform EdgeAA

DP

EdgeAA

Scaleform Player Flash CPU CPU

Scaleform CTRL+Y

60 CPU

release debug

Scaleform debug Scaleform

3.4

Scaleform 4.1 Scaleform\GFx SDK 4.1\Apps\Tutorial SLN

Scaleform→ GFx SDK 4.1→ Tutorial Windows for Visual Studio 2005

Visual Studio 2008 “Debug”

2 ShadowVolume

3.5 Scaleform

Scaleform Visual Studio Tutorial
$(GFXSDK) SDK
C:\Program Files\Scaleform\GFx SDK 4.2\
$(GFXSDK) “Msvc90
Visual Studio 2008 Visual Studio 2010 Msvc90 Msvc10

Scaleform
$(GFXSDK)\Src
$(GFXSDK)\Include
Visual Studio “Additional Include Directories”

AS3 AS3
“GFx/AS3/AS3_Global.h” AS3
Scaleform LITE

$(DXSDK_DIR)\Lib\x86
$(GFXSDK)\3rdParty\expat-2.1.0\lib
$(GFXSDK)\Lib\$(PlatformName)\Msvc90\Debug_Static\
$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\$(PlatformName)\Msvc90\Debug
$(GFXSDK)\3rdParty\jpeg-8d\Lib\$(PlatformName)\Msvc90\Debug

“Additional Library Directories”
"$(DXSDK_DIR)\Lib\x86";
"$(GFXSDK)\3rdParty\expat-2.1.0\lib";
"$(GFXSDK)\Lib\$(PlatformName)\Msvc90\debug";
"$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\$(PlatformName)\Msvc90\Debug";
"$(GFXSDK)\3rdParty\jpeg-8d\Lib\$(PlatformName)\Msvc90\Debug"
Msvc90 Visual Studio

$(DXSDK_DIR)\Lib\x86
$(GFXSDK)\3rdParty\expat-2.1.0\lib
$(GFXSDK)\Lib\$(PlatformName)\Msvc90\Release\
$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\$(PlatformName)\Msvc90\Release
$(GFXSDK)\3rdParty\jpeg-8d\Lib\$(PlatformName)\Msvc90\Release

Additional Library Directories”

"$(DXSDK_DIR)\Lib\x86";
"$(GFXSDK)\3rdParty\expat-2.1.0\lib";
"$(GFXSDK)\Lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\jpeg-8d\Lib\$(PlatformName)\Msvc90\Release"
"$(GFXSDK)3rdParty\libpng\Lib\$(PlatformName)\Msvc90\Release”

Msvc90 Visual Studio

Scaleform

libgfx.lib
libAS2.lib
libAS3.lib
libgfxexpat.lib
libjpeg.lib
zlib.lib
libpng.lib
libgfxrender_d3d9.lib
libgfxsound_fmod.lib

.vcproj

Scaleform include Tutorial\Section3.5

3.5.1 AS3_Global.h and Obj\AS3_Obj_Global.xxx

AS3_Global.h Obj\AS3_Obj_global.xxx

AS3_Obj_Global.xxx “ ”ActionScript 3 swf

“ ”(script) GlobalObjectCPP C++

Scaleform VM

AS3_Global.h ClassRegistrationTable

AS3 C++

ClassRegistrationTable

AS3_Global.h

ClassRegistrationTable include

ClassRegistrationTable AS3_Global.h

AS3 VM

“ ”

4

Scaleform 3D Unreal® Engine 3, Gamebryo™, Bigworld®, Hero Engine™, Touchdown Jupiter Engine™, CryENGINE™, Trinigy Vision™
Scaleform

Scaleform DirectX DirectX ShadowVolume SDK
DirectX
DXUT 2D 3D

Scaleform Flash Scaleform DXUT

DXUT
Win32 DirectX Scaleform SDK GFxPlayerTiny.cpp

4.1 Flash

Flash 3D GFx::Loader
Flash GFx::MovieDef Flash GFx::Movie
Render::Renderer2D Render::HAL DirectX
Scaleform API
/

ShadowVolume 4.1

#1:

ShadowVolume.cpp

```
#include "GFx_Kernel.h"
#include "GFx.h"
#include "GFx_Renderer_D3D9.h"
```

Scaleform GFxTutorial
Scaleform

Scaleform

```
//                GFx::Loader
Loader           gfxLoader;

//    SWF/GFx           GFx::MovieDef
Ptr<MovieDef>   pUIMovieDef;

//                    GFx::Movie
Ptr<Movie>  pUIMovie;

// Renderer
Ptr<Render::D3D9::HAL>  pRenderHAL;
Ptr<Render::Renderer2D> pRenderer;
MovieDisplayHandle      hMovieDisplay;
```

#2 GFx::System

Scaleform GFx::System Scaleform WinMain

```
//                GFx::System
GFx::System          gfxInit;
```

GFx::System Scaleform Scaleform
WinMain GFx::System Scaleform
Tutorial
GFx::System

GFx::System
GFxTutorial GFx::System

GFx::System
GFx::System::Init() GFx::System::Destroy()

#3:

Scaleform WinMain InitApp()
InitApp()

```
gfx = new GFxTutorial();
assert(gfx != NULL);
```

```
if(!gfx->InitGFx())
      assert(0);
```

GFxTutorial GFx::Loader GFx::Loader
SWF/GFx SWF/GFx
SWF/GFx GFx::Loader
GFx::Log,

GFxTutorial::InitGFx() GFx::Loader GFx::Loader
SetLog Scaleform
Scaleform PlayerLog
GFx::Log

```
//          –Scaleform
//
gfxLoader->SetLog(Ptr<Log>(*new GFxPlayerLog()));
```

GFx::Loader GFx::FileOpener
GFx::FileOpener

```
//
Ptr<FileOpener> pfileOpener = *new FileOpener;
gfxLoader->SetFileOpener(pfileOpener);
```

Render::HAL Scaleform D3D9
D3D
Scaleform HAL::InitHAL Render HAL
IDirect3Device9 Scaleform DX9
UI

```
pRenderHAL = *new Render::D3D9::HAL();
if (!(pRenderer = *new Render::Renderer2D(pRenderHAL.GetPtr())))
     return false;
```

Scaleform Render::HAL Render::D3D9::HAL
Scaleform

#4 Flash

GFx::Loader GFx::MovieDef
GFx::MovieDef

```
//
pUIMovieDef = *gfxLoader.CreateMovie(UIMOVIE_FILENAME,
                                     Loader::LoadKeepBindData |
                                     Loader::LoadWaitFrame1, 0);
```

LoadKeepBindData D3D

LoadWaitFrame1 CreateMovie
GFx::ThreadedTaskManager

Memory System Overview

#5

GFx::MovieDef GFx::Movie GFx::Movie

```
pUIMovie = *pUIMovieDef->CreateInstance(true, 0, NULL);
assert(pUIMovie.getPtr() != NULL);
```

CreateInstance false
Flash
Memory System
Overview

Advance() false CreateInstance

```
//
pUIMovie->Advance(0.0f, 0, true);

//
MovieLastTime = timeGetTime();
```

#8

Scaleform GFxTutorial
WinMain GFxTutorial

```
delete gfx;
gfx = NULL;
```

DirectX 9 Scaleform
InitHAL() ShutdownHAL()

DirectX 9 InitHAL D3DPOOL_DEFAULT
InitHAL D3DPOOL_DEFAULT

ShutdownHAL D3DPOOL_DEFAULT DXUT
ShadowVolume DXUT OnResetDevice D3DPOOL_DEFAULT
OnLostDevice GFxTutorial::OnLostDevice
ShutdownHAL GFxTutorial::OnResetDevice InitHAL

InitHAL ShutdownHAL
D3DPOOL_DEFAULT

#9

D3D
GFxTutorial::OnResetDevice

```
//
RECT windowRect = DXUTGetWindowClientRect();
DWORD windowWidth = windowRect.right - windowRect.left;
DWORD windowHeight = windowRect.bottom - windowRect.top;
pUIMovie->SetViewport(windowWidth, windowHeight, 0, 0,
                      windowWidth, windowHeight);
```

SetViewport PC
Scaleform

OpenGL

Scaleform Flash 4.2

#10 DirectX

ShadowVolume OnFrameRender() D3D
BeginScene() EndScene() EndScene()
GFxTutorial::AdvanceAndRender()

```
void AdvanceAndRender(void)
{
      DWORD mtime = timeGetTime();
      float deltaTime = ((float)(mtime - MovieLastTime)) / 1000.0f;
      MovieLastTime = mtime;

      pUIMovie->Advance(deltaTime, 0);
      pRenderer->BeginFrame();

      if (hMovieDisplay.NextCapture(pRenderer->GetContextNotify()))
      {
          pRenderer->Display(hMovieDisplay);
      }

      pRenderer->EndFrame();
}
```

Advance() deltaTime
GFx::Movie::Advance

#11

Render::Renderer2D::Display DirectX D3D
D3D
Render::Renderer2D::Display D3D

Scaleform
DX9

```
    //     Scaleform     DirectX
    g_pStateBlock->Capture();

    //
    gfx->AdvanceAndRender();

    //     DirectX
    g_pStateBlock->Apply();
```

#12 UI

DXUT 4.1

ShadowVolume.cpp

DXUT

```
    // Disable default UI
    ...
```

DirectX Flash

3 Scaleform Flash ShadowVolum

4.2

GFx::Movie::SetViewport Flash
Flash Scaleform

- .
-

4:3 Scaleform

800×600 1600×1200
Scaleform
Scaleform

GFx::Movie::SetViewScaleMode
GFxTutorial::InitGFx() GFx::Movie

```
pUIMovie->SetViewScaleMode(Movie::SM_ShowAll);
```

SetViewScaleMode

SM_NoScale	Flash
SM_ShowAll	
SM_ExactFit	
SM_NoBorder	

SetViewAlignment SM_NoScale
SM_ShowAll SetViewAlignment

```
pUIMovie->SetViewAlignment(Movie::Align_CenterRight);
```

SetViewScaleMode SetViewAlignment

SetViewAlignment SetViewScaleMode SM_NoScale
Scaleform
d3d9guide.fla

ActionScript C++ SetViewScaleMode SetViewAlignment ActionScript Stage (State.scaleMode, Stage.align)

4.3

ShadowVolume Scaleform Flash Flash

GFx::Movie::HandleEvent GFx::Event GFx::Movie

4.3.1

ShadowVolume MsgProc Win32 GFxTutorial::ProcessEvent Scaleform WM_MOUSEMOVE, WM_LBUTTONDOWN WM_LBUTTONUP

```
void ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam, LPARAM lParam,
                  bool *pbNoFurtherProcessing)
{
      int mx = LOWORD(lParam), my = HIWORD(lParam);
      if(pUIMovie)
      {
            if(uMsg == WM_MOUSEMOVE)
            {
                  MouseEvent mevent(GFx::Event::MouseMove, 0, mx, my);
                  pUIMovie->HandleEvent(mevent);
            }
            else if(pMovieButton && uMsg == WM_LBUTTONDOWN)
            {
                  ::SetCapture(hWnd);
                  MouseEvent mevent(GFx::Event::MouseDown, 0, mx, my);
                  pUIMovie->HandleEvent(mevent);
            }
            else if(pMovieButton && uMsg == WM_LBUTTONUP)
            {
                  ::ReleaseCapture();
                  MouseEvent mevent(GFx::Event::MouseUp, 0, mx, my);
                  pUIMovie->HandleEvent(mevent);
            }
      }
}
```

Scaleform

#1:

```
pMovie->SetViewport(screen_width, screen_height, 0, 0, screen_width,
                                                    screen_height, 0);
```

(0, 0)
Scaleform

#2

```
pMovie->SetViewport(screen_width, screen_height, 0, 0, screen_width / 4,
                                                screen_height / 4, 0);
```

HandleEvent
Scaleform

#3

```
movie_width = screen_width / 6;
movie_height = screen_height / 6;
pMovie->SetViewport(screen_width, screen_height,
                         screen_width / 2 – movie_width / 2,
                        screen_height / 2 – movie_height / 2,
                        movie_width, movie_height);
```

(0,0) (screen_width / 2 – movie_width / 2, screen_height / 2 – movie_height / 2)

Flash GFx::Movie::SetViewAlignment
GFx::Movie::SetViewport SetViewAlignment SetViewScaleMode
Scaleform

4.3.2

GFx::Movie:HandleEvent GFx::KeyEvent GFx::CharEvent:

```
KeyEvent(EventType eventType = None,
                  Key::Code code = Key::None,
                  UByte asciiCode = 0,
```

```
                    UInt32 wcharCode = 0,
                    UInt8 keyboardIndex = 0)

    CharEvent(UInt32 wcharCode, UInt8 keyboardIndex = 0)
```

GFx::KeyEvent GFx::CharEvent ASCII Windows GFx::CharEvent WM_CHAR GFx::KeyEvents WM_SYSKEYDOWN, WM_SYSKEYUP, WM_KEYDOWN WM_KEYUP

- ‘c’ SHIFT : GFx::KeyEvent WM_KEYDOWN
  - ‘c’
  -
  - SHIFT
- GFx::CharEvent WM_CHAR “ ” ASCII ‘C’ GFx
- ‘c’ GFx::KeyEvent WM_KEYUP GFx::CharEvent
- F5 GFx::KeyEvent F5 ASCII GFx::CharEvent

GFx::KeyEvent GFx_Event.h Flash GFxPlayerTiny.cpp Scaleform Player Windows Flash ProcessKeyEvent 3D

```
    void ProcessKeyEvent(Movie *pMovie, unsigned uMsg, WPARAM wParam, LPARAM
    lParam)
```

Windows WndProc WM_CHAR, WM_SYSKEYDOWN, WM_SYSKEYUP, WM_KEYDOWN WM_KEYUP Scaleform pMovie

GFx::KeyEvent GFx::CharEvent GFx::CharEvents ASCII Unicode IME . IME IME GFx::KeyEvent Page Up Page Down

4.3

```
{
	::ReleaseCapture();
	MouseEvent mevent(GFx::Event::MouseUp, 0, (float)mx, (float)my);
	pUIMovie->HandleEvent(mevent);
	processedMouseEvent = true;
}

if(processedMouseEvent && pUIMovie->HitTest((float)mx, (float)my,
	Movie::HitTest_Shapes))
	*pbNoFurtherProcessing = true;
```

4.3.4

“Change Mesh”

4.3.2 W,S,A,D Q 3D Scaleform 3D

6.1.2

C++

5

5.1

Scaleform

Flash

TTF

Windows Linux Scaleform

https://gameware.autodesk.com/scaleform/developer/?action=doc

5000

Scaleform

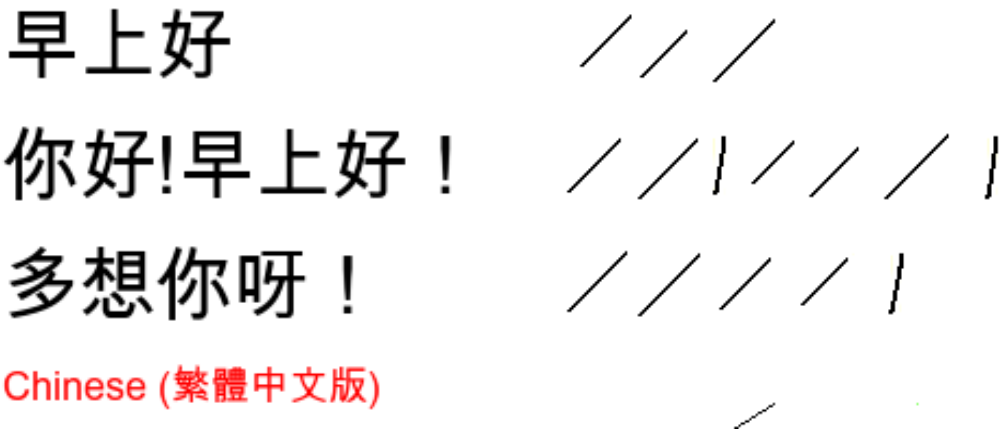
Scaleform

Scaleform TTF

Scaleform

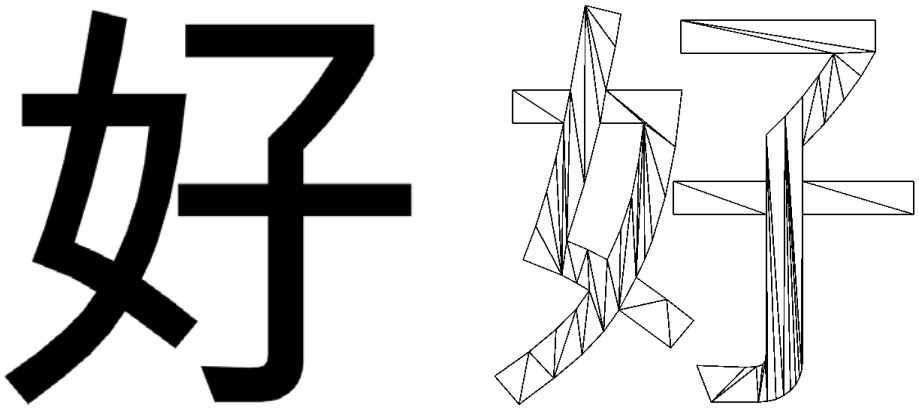
Bin\Data\AS2\Samples\FontConfig sample.swf

Scaleform Player D3D9



3a **4b**

CTRL+W



4c **4d**

4c 4d Scaleform

Flash

gfx_2.1_texteffects_sample.zip

Isotropic shadow with offset
Text Shadow
文本陰影
Anisotropic glow, BlurX=2, BlurY=7
Text Glow 1
文本煥發 1
Isotropic glow, BlurX=BlurY=4, Knockout
TEXT GLOW 2
文本煥發 2
Rotating glow
Text Glow 3
文本煥發 3
Changing blurriness
Text Glow 4
文本煥發 4

5

6 C++ Flash

Flash

Flash

Scaleform C++

Scaleform

C++

6.1 C++

Scaleform C++ FSCommand

ExternalInterface FSCommand ExternalInterface GFx::Loader C++

FSCommand fscommand

ExternalInterface flash.external.ExternalInterface.call

GFx::Value 6.1.2

FSCommands ExternalInterface

fscommands gfxexport SWF

fscommands –fstree –fslist –fsparams

ExternalInterface fscommands

6.1.1 FSCommand

fscommand

```
fscommand("setMode", "2");
```

fscommand

ExternalInterface

GFx FSCommand GFx::FSCommandHandler

FSCommand GFx::Loader GFx::Movie

GFx::Movie fscommand Scaleform PlayerTiny

(“FxPlayerFSCommandHandler”) ShadowVolume

6.1

GFx::FSCommandHandler

```
class OurFSCommandHandler : public FSCommandHandler
{
    public:
    virtual void Callback(Movie* pmovie,
                          const char* pcommand, const char* parg)
    {
        GFxPrintf("FSCommand: %s, Args: %s", pcommand, parg);
    }
};
```

Callback fscommand
fscommand

GFxTutorial::InitGFx() GFx::Loader

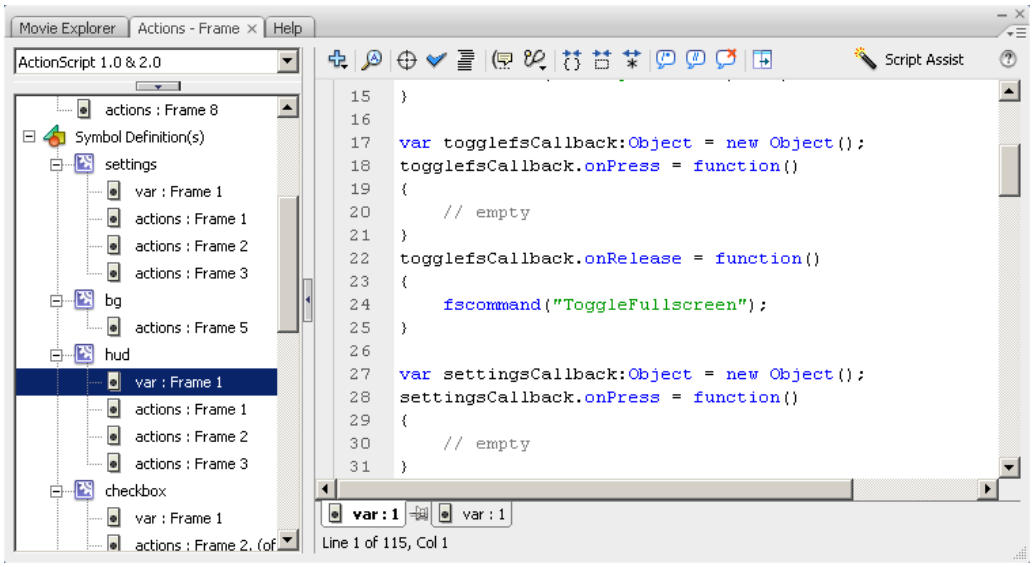
```
//      FSCommand
Ptr<FSCommandHandler> pcommandHandler = *new OurFSCommandHandler;
gfxLoader->SetFSCommandHandler(pcommandHandler);
```

GFx::Loader GFx::Movie SetFSCommandHandler

fscommand ShadowVolume
“Toggle Fullscreen”

```
FSCommand: ToggleFullscreen, Args:
```

Flash Studio d3d9guide.swf (F9)
ActionScript block for Symbol Definition(s) → hud → var : Frame 1
fscommand



6 fscommand()

fscommand(“ToggleFullscreen”) fscommand(“ToggleFullscreen”, this.UI._visible)
C++ this.UI._visible flash (CTRL+ALT+Shift+S) d3d9guide.swf

FSCommand ShadowVolume
fscommand

```
if(strcmp(pcommand, "ToggleFullscreen") == 0)
      doToggleFullscreen = true;
```

DXUT OnFrameMove OnFrameMove

```
if(doToggleFullscreen)
{
      doToggleFullscreen = false;
      DXUTToggleFullScreen();
}
```

fscommand doToggleFullscreen DXUT
OnFrameMove

6.1.2 ExternalInterface

Flash ExternalInterface fscommand

ExternalInterface fscommand

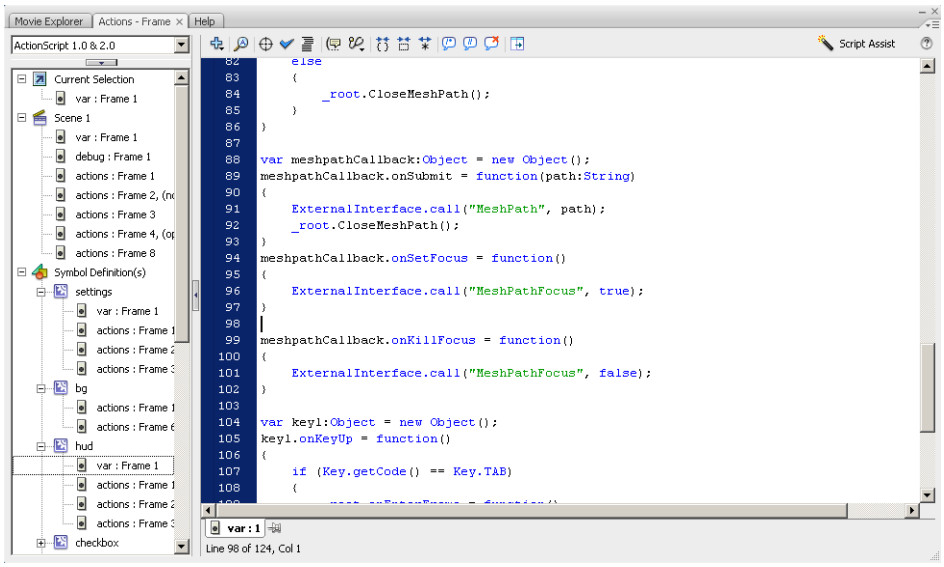
```
class OurExternalInterfaceHandler : public ExternalInterface
{
  public:
      virtual void Callback(Movie* pmovieView,
                            const char* methodName,
                            const Value* args,
                            unsigned argCount)
      {
            GFxPrintf("ExternalInterface: %s, %d args: ",
                                          methodName, argCount);
            for(unsigned i = 0; i < argCount; i++)
            {
                  switch(args[i].GetType())
                  {
                    case Value::VT_Null:
                        GFxPrintf("NULL");
                        break;
                    case Value::VT_Boolean:
                        GFxPrintf("%s", args[i].GetBool()? "true" : "false");
                        break;
                    case Value::VT_Number:
                        GFxPrintf("%3.3f", args[i].GetNumber());
                        break;
                    case Value::VT_String:
                        GFxPrintf("%s", args[i].GetString());
                        break;
                    default:;
                        GFxPrintf("unknown");
                        break;
                  }
                  GFxPrintf("%s", (i == argCount - 1) ? "" : ", ");
            }
            GFxPrintf("\n");
      }
};
```

GFx::Loader

```
Ptr<ExternalInterface> pEIHandler = *new OurExternalInterfaceHandler;
gfxLoader.SetExternalInterface(pEIHandler);
```

F9

ActionScript for Symbol Definitions() → hud → var : Frame 1




```
    else
    {
        _root.CloseMeshPath();
    }
}

var meshpathCallback:Object = new Object();
meshpathCallback.onSubmit = function(path:String)
{
    ExternalInterface.call("MeshPath", path);
    _root.CloseMeshPath();
}
meshpathCallback.onSetFocus = function()
{
    ExternalInterface.call("MeshPathFocus", true);
}

meshpathCallback.onKillFocus = function()
{
    ExternalInterface.call("MeshPathFocus", false);
}

var key1:Object = new Object();
key1.onKeyUp = function()
{
    if (Key.getCode() == Key.TAB)
    {
```

7 MeshPathFocus

(true false)

"MeshPathFocus"

```
ExternalInterface: MeshPathFocus, 1 args: false
Focus change:
        _level0.hud.text_MeshPath.field => null
ExternalInterface: MeshPathFocus, 1 args: true
Focus change:
        null => _level0.hud.text_MeshPath.field
```

GFxTutorial

```
f(strcmp(methodName, "MeshPathFocus") == 0 && argCount == 1)
      {
            if(args[0].GetType() == Value::VT_Boolean)
                  gfx->SetTextboxFocus(args[0].GetBool());
      }
```

GFxTutorial::ProcessEvent

3D

```
if(uMsg == WM_SYSKEYDOWN || uMsg == WM_SYSKEYUP ||
   uMsg == WM_KEYDOWN    || uMsg == WM_KEYUP    ||
   uMsg == WM_CHAR)
{
      if(textboxHasFocus || wParam == 32 || wParam == 9)
      {
            ProcessKeyEvent(pUIMovie, uMsg, wParam, lParam);
            *pbNoFurtherProcessing = true;
      }
}
```

(ASCII 32) tab(ASCII 9) “Toggle UI” “Settings”

“Change Mesh”

OurExternalInterfaceHandler::Callback

```
static bool doChangeMesh = false;
static wchar_t changeMeshFilename[MAX_PATH] = L"";

...

if(strcmp(methodName, "MeshPath") == 0 && argCount == 1)
{
      doChangeMesh = true;
      const char *filename = args[0].GetString();
      MultiByteToWideChar(CP_ACP, MB_PRECOMPOSED, filename, -1,
            changeMeshFilename, _countof(changeMeshFilename));
}
```

DXUT OnFrameMove DXUT

6.2 C++

6.1 C++ Scaleform

C++ Scaleform C++

6.2.1

Scaleform GetVariable SetVariable 6.2
HUD (fxplayer.swf) Scaleform Player
F5

```
void GFxTutorial::ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam,
                               LPARAM lParam, bool *pbNoFurtherProcessing)
{
      int mx = LOWORD(lParam), my = HIWORD(lParam);

      if(pHUDMovie && uMsg == WM_KEYDOWN)
      {
            if(wParam == VK_F5)
            {
                  int counter = (int)pHUDMovie-
                                        >GetVariableDouble("_root.counter");
                  counter++;
                  pHUDMovie->SetVariable("_root.counter",
                                          Value((double)counter));
                  char str[256];
                  sprintf_s(str, "testing! counter = %d", counter);
                  pHUDMovie->SetVariable("_root.MessageText.text", str);
            }
      }
      ...
```

GetVariableDouble _root.counter GetVariableDouble
SetVariable _root.counter
GFx::Movie GetVariable SetVariable

fxplayer Flash MessageText
HUDText _root.counter SetVariable

Flash TextField.text GFx::Movie::Invoke
6.2.2
Scaleform

8 SetVariable HUD

GFx::Value

SetVariable GFx::Movie::SetVarType

_root.mytextfield 3 SetVariable(“_root.mytextfield.text”, “testing”, SV_Normal) 1 SV_Sticky 3 _root.mytextfield.text C++

SetVariableArray Flash GFxTutorial::InitGFx() _root.SceneData

```
//
Value sceneData[3];
sceneData[0].SetString("Scene with shadow");
sceneData[1].SetString("Show shadow volume");
sceneData[2].SetString("Shadow volume complexity");
pUIMovie->SetVariableArraySize("_root.SceneData", 3);
pUIMovie->SetVariableArray(Movie::SA_Value,
      "_root.SceneData", 0, sceneData, 3);
```

6.1
ShadowVolume

C++
C++

6.2.2

GFx::Movie::Invoke
UI

GFx::Movie::Invoke “Change Mesh” F6 F7
SetVariable
SetVariable Invoke ActionScript

GFx::Movie::Invoke WM_CHAR openMeshPath

```
...

else if (wParam == VK_F6)
{
      bool retval = pHUDMovie->Invoke("_root.OpenMeshPath", "");
      GFxPrintf("_root.OpenMeshPath returns '%d'\n", (int) retval);
}
else if (wParam == VK_F7)
{
      const char *retval = pHUDMovie->Invoke("_root.CloseMeshPath", "");
      GFxPrintf("_root.CloseMeshPath returns '%d'\n", (int) retval);
}

...
```

openMeshPath 1 1
Scaleform

GFx::Movie::Advance GFx::MovieDef::CreateInstance
initFirstFrame true 1

printf GFx::Value
InvokeArgs va_list Invoke
InvokeArgs printf vprintf

6.3 *Flash*

C++ SWF
C++

MMOG HUD HUD
HUD
HUD

SWF C++
GFx::Movie

loadMovie unloadMovie
SWF GFx::Movie SWF GFx::Movie
C++

MMOG _global.inventory SWF
SWF unloadMovie
_global.inventory

ActionScript container.swf
container.swf ActionScripts
MovieClipLoader

```
var mclLoader:MovieClipLoader = new MovieClipLoader();
```

ActionScript Flash

```
////
//

function LoadFlashLevel(url, level)
{
```

```
    trace("LoadFlashLevel(" + url + ", " + level + ")\n");
    mclLoader.loadClip(url, level);
}

//
//

function LoadFlash(url, objectName)
{
    trace("LoadFlashLevel(" + url + ", " + objectName + ")\n");
    var container:MovieClip = createEmptyMovieClip(objectName,
                                                   getNextHighestDepth());
    mclLoader.loadClip(url, container);
}
```

level

Z UI

LoadMovieLevel Z

_levelN.variableName level6.counter

LoadMovie

root.invⅧ2eLryaB6BWindow.widget1.cu-5aB6Ber

Flash root ro12211(t)ⅧJ/C2_0 1 Tf0 Tc 0 Tw 2.054 0 T

root root root

oeLScript Adobe Flash

GFx::Movie

tainer.fla

LoadMovie LoadMovieLevel loadClip

MovieClipLoader Container.fla
ActionScript ExternalInterface C++

```
//
//   1.
//   2.
//   3.
//   4.
//
//
//          LoadMovie()
//
//
//
//         ExternalInterface        C++
//       ActionScript

var mclListener:Object = new Object();

mclListener.onLoadError = function(target_mc:MovieClip, errorCode:String,
                                   status:Number)
{
    trace("Error loading image: " + errorCode + " [" + status + "]");
};

mclListener.onLoadStart = function(target_mc:MovieClip) : Void
{
    trace("onLoadStart: " + target_mc);
};

mclListener.onLoadProgress = function(target_mc:MovieClip,
                                      numBytesLoaded:Number,
                                      numBytesTotal:Number) : Void
{
    var numPercentLoaded:Number = numBytesLoaded / numBytesTotal * 100;
    trace("onLoadProgress: " + target_mc + " is " +
          numPercentLoaded + "% loaded");
};

mclListener.onLoadComplete = function(target_mc:MovieClip,
                                      status:Number) : Void
```

```
    {
        trace("onLoadComplete: " + target_mc);
    };

    // Register the listener with the MovieClipLoader
    mclLoader.addListener(mclListener);
```

Tutorial\section6.2 container.swf d3d9guide.swf fxplayer.swf
F8 HUD F9 UI

```
void GFxTutorial::ProcessEvent(HWND hWnd, unsigned uMsg,
                               WPARAM wParam, LPARAM lParam,
                               bool *pbNoFurtherProcessing)
{
    ...

  else if (wParam == VK_F8)
  {
    bool retval = pShellMovie->Invoke("_root.LoadFlashLevel",
                                      "%s, %d", "fxplayer.swf", 5);
    GFxPrintf("_root.LoadFlash returns '%d'\n", (int) retval);
  }
  else if (wParam == VK_F9)
  {
    bool retval = pShellMovie->Invoke("_root.LoadFlashLevel",
                                      "%s, %d", "d3d9guide.swf", 6);
    GFxPrintf("_root.LoadFlash returns '%d'\n", (int) retval);
  }
  else if (wParam == VK_F2)
  {
    bool retval = pContainerMovie->Invoke("_root.UnloadFlashLevel",
                                          "%d", 5);
    GFxPrintf("_root.UnloadFlash returns '%d'\n", (int) retval);
  }
  else if (wParam == VK_F3)
  {
    bool retval = pContainerMovie->Invoke("_root.UnloadFlashLevel",
                                          "%d", 6);
    GFxPrintf("_root.UnloadFlash returns '%d'\n", (int) retval);
  }
}
```

HUD 5 6 Flash z
GFx::Movie Container.swf
0

_level _level6 HUD
_level5.MessageText.text
_global Flash _global.counter
_global
_global _level, _global
Adobe Flash

F8 HUD F9
SWF
Scaleform 8 Scaleform

F8 HUD F5 6.2 SetVariable
_root.counter HUD _level5

```
void GFxTutorial::ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam, LPARAM
                               lParam, bool *pbNoFurtherProcessing)
{
      int mx = LOWORD(lParam), my = HIWORD(lParam);

      if(pHUDMovie && uMsg == WM_KEYDOWN)
      {
            if(wParam == VK_F5)
            {
                  int counter = (int)pHUDMovie->
                                       GetVariableDouble("_level5.counter");
                  counter++;
                  pHUDMovie->SetVariable("_level5.counter",
                                       Value((double)counter));
                  char str[256];
                  sprintf_s(str, "testing! counter = %d", counter);
                  pHUDMovie->SetVariable("_level5.MessageText.text", str);
            }
      }

      ...
```

_level5.counter d3d9guide.swf 6
C++

F8 HUD F5 F2 HUD F8 HUD
F5 HUD

_level5 _level5.counter _global.counter
_global

7

3D

“ ” Flash Studio SWF SWF

UI

UI

1. cell.x

(cellwall.jpg)

UI

cellwallRT.jpg MeshMaterialList

```
Material {
1.000000;1.000000;1.000000;1.000000;;
40.000000;
1.000000;1.000000;1.000000;;
0.000000;0.000000;0.000000;;
```

```
TextureFilename {
"cellwallRT.jpg";
}
```

2. cell.x DXUT

   DXUT

   CreateTexture (Render Texture) DXSDK

   DXUT

```
pd3dDevice->CreateTexture(rtw,rth,0,
D3DUSAGE_RENDERTARGET|D3DUSAGE_AUTOGENMIPMAP, D3DFMT_A8R8G8B8,
            D3DPOOL_DEFAULT, &g_pRenderTex, 0);

g_Background[0].m_pTextures[BACK_WALL] = g_pRenderTex;
ptex->Release();
```

3. AdvanceAndRender GFx

   a.
   ```
   pd3dDevice->GetRenderTarget(0, &poldSurface);
   pd3dDevice->GetDepthStencilSurface(&poldDepthSurface);
   ```
   b.
   ```
   ptex->GetSurfaceLevel(0, &psurface);
   HRESULThr = pd3dDevice->SetRenderTarget(0, psurface );
   ```
   c. Display a

8 GFxExport

SWF SWF
SWF SWF
SWF

GFxExport SWF
DDS DXT

GFxExport

Scaleform GFxExport
d3d9guide.swf fxplayer.swf Scaleform

```
gfxexport -i DDS -c d3d9guide.swf
gfxexport -i DDS -c fxplayer.swf
```

gfxexport.exe C:\Program Files\Scaleform\$(GFXSDK)
convert.bat 7

-i DDS DDS DirectX DXT
DXT

-c Scaleform
DXT

-share_images

ShadowVolume 8 Scaleform
GFx::Loader::CreateMovie

9

Scaleform

- Flash Scaleform Player SWF SDK Gamebryo 3
- FAQs
- Scale9 Scale9Grid Overview
- Scaleform SWF Flash GFx::FileOpener
- GFx::ImageCreator
- GFx::Translator